

CHEMISTRY

OVERVIEW

The Department of Chemistry offers Bachelor of Science and Bachelor of Arts degrees in Chemistry in the College of Liberal Arts and Sciences.

Graduates holding the **B.S. degree in Chemistry** qualify in many fields as teachers of Chemistry, supervisors in industry, technical sales personnel, and research chemists in federal, state, municipal, academic, or industrial laboratories. Students with high scholastic standing often continue with graduate work, where they may explore more thoroughly the specialized areas of chemistry in which they are interested. The B.S. degree is recommended for students who wish to pursue graduate studies or a research career in Chemistry.

The **B.A. degree in Chemistry** is useful for students who want to become teachers, attend professional school (e.g., medical, veterinarian, law) or want to pursue non-traditional chemistry careers, such as Forensic Science Technician, Environmental Policy Analyst, Science Communicator, Technical Writer, or a Laboratory Technician. Students enrolled in the B.A. in Chemistry complete pathways geared toward specific career types. These pathways also make additional majors and minors in other disciplines extremely feasible.

Student Learning Outcomes

Upon graduation students should:

- Have firm foundations in the fundamentals and application of current chemical theories.
- Be able to design, carry out, record, and analyze the results of chemical experiments.
- Use modern instrumentation and classical techniques to identify and solve chemical problems as well as explore new areas of research.
- Communicate the results of their work to chemists, as well as non-chemists.
- Understand the ethical and environmental dimensions of problems and issues facing chemists.
- Follow the proper procedures and regulations for safe storage, labeling, use of chemicals, and disposal of chemicals.
- Be able to use chemical literature (perform searches for, read, evaluate, and appropriately cite publications).
- Be skilled in problem solving, critical thinking, and analytical reasoning.

These skills may be applied to careers in education and industry; in professions such as law, medicine, environmental sciences, and forensic sciences. The Chemistry, B.S. is approved by the American Chemical Society (ACS). Students who complete the program obtain an ACS certified

baccalaureate degree provided they also take one Biochemistry course, typically BBMB 3010 Survey of Biochemistry, BBMB 3160 Principles of Biochemistry, or BBMB 4040 Biochemistry I and BBMB 4050 Biochemistry II.

DEGREE REQUIREMENTS

All Chemistry students complete required classes in Chemistry, Physics, and Math, and additional credits of Advanced Coursework depending on the degree type. Students choose a pathway based on their career goals: Chemistry (B.S. or B.A.), Teacher Preparation (with B.A. or B.S.), or chemistry applications (e.g., pre-Med) via career course choices in the B.A.

Chemistry, B.S.

Undergraduate students seeking the Bachelor of Science (B.S.) degree in Chemistry complete 67-69 credits in required chemistry, math, and physics classes, and have the following courses in their degree programs as minimum requirements.

Chemistry Core Courses (B.S.)

CHEM 1770	General Chemistry I	4
CHEM 1770L	Laboratory in General Chemistry I	1
CHEM 1780	General Chemistry II	3
CHEM 2110	Quantitative and Environmental Analysis	2
CHEM 2110L	Quantitative and Environmental Analysis Laboratory	2
CHEM 3010	Inorganic Chemistry ^S	2
CHEM 3160	Instrumental Methods of Chemical Analysis ^F	2
CHEM 3160L	Instrumental Analysis Laboratory ^F	2
CHEM 3220L	Laboratory in Physical Chemistry ^S	3
CHEM 3240	Introductory Quantum Mechanics	3
CHEM 3250	Chemical Thermodynamics	3
CHEM 3310	Organic Chemistry I	3
CHEM 3320	Organic Chemistry II	3
CHEM 3330L	Laboratory in Organic Chemistry I (for Chemistry and Biochemistry Majors) ^F	2
CHEM 3340L	Laboratory in Organic Chemistry II (for Chemistry and Biochemistry Majors) ^S	2
CHEM 4020	Advanced Inorganic Chemistry	3
CHEM 4010L	Inorganic Chemistry Laboratory	1
Advanced Chemistry Choice 1 ²		2-3
Advanced Chemistry Choice 2 ²		2-3

[CHEM 1670/CHEM 1670L (5 cr.)] may be substituted for [CHEM 1770/CHEM 1770L (5 cr.)] only if the Chemistry major has been declared after receiving credit for [CHEM 1670/CHEM 1670L

[CHEM 1770/CHEM1770N/CHEM 178 (8 cr.)) may be substituted with [CHEM 2010/CHEM 2010L (6 cr.))]^{1,F}

[CHEM 3310L/CHEM 3320L] may be substituted for [CHEM 3330L/CHEM 3340L]. However, this substitution may result in a program which is deficient in the laboratory experience recommended by the American Chemical Society.

Total Credits

45-47

¹ Advanced high school chemistry and strong algebra skills are necessary for success in CHEM 2010. Math ACT of 24 or greater is strongly recommended.

² The completion of two courses (minimum of 2 credits each) is mandatory to meet this requirement. See below for details on this requirement and for a list of accepted courses.

^F Offered only in Fall

^S Offered only in Spring

Additional Chemistry Courses (B.S.)

CHEM 1010A	Chemistry Learning Community Orientation: On-Campus Orientation ^F	1
CHEM 1010B	Chemistry Learning Community Orientation: Professional Development Opportunities ^S	1
CHEM 1100	Cutting-Edge Chemistry: Research and Career Opportunities ^F	1

CHEM 1010A and CHEM 1010B are required for students in the Chemistry Learning Community

All students majoring in Chemistry are strongly encouraged to take CHEM 1100

^F Offered only in Fall

^S Offered only in Spring

Math and Physics Core Courses (B.S.)

MATH 1650	Calculus I	4
MATH 1660	Calculus II	4
MATH 2650	Calculus III	4
PHYS 2310	Introduction to Classical Physics I	4
PHYS 2310L	Introduction to Classical Physics I Laboratory	1
PHYS 2320	Introduction to Classical Physics II ^S	4
PHYS 2320L	Introduction to Classical Physics II Laboratory ^S	1

Total Credits

22

Advanced Chemistry Choices (B.S.)

The completion of two courses (minimum of 2 credits each) is mandatory within the Chemistry BS degree. In addition to Advanced Chemistry Courses (4000 and 5000 level), certain courses in Biochemistry-Biophysics,

Chemical Engineering, Computer Science, Mathematics, and Materials Science and Engineering are acceptable.

Students may assume courses in this list will be accepted without petition but should petition through their adviser for the acceptance of any other course outside this list offered by a department other than chemistry. Lab courses associated with lectures (e.g., FS HN 3110L) are not considered as advanced chemistry courses.

Up to three credits in undergraduate research (CHEM 3990/4990/4900) can be counted as one of the two advanced chemistry courses. CHEM 5500 and CHEM 5550 may not be used to satisfy the Advanced Chemistry requirement.

List of Approved Advanced Chemistry Credits

CHEM 3260	Chemical Kinetics	1
CHEM 3980	Cooperative Education	R
CHEM 3990	Undergraduate Research	1-30
CHEM 4900	Independent Study	1-30
CHEM 4980	Cooperative Education	R
CHEM 4990	Senior Research	2-3
CHEM 5010L	Inorganic Preparations	1
CHEM 5020	Advanced Inorganic Chemistry	3
CHEM 5050	Physical Inorganic Chemistry	3
CHEM 5110	Advanced Analytical Chemistry	3
CHEM 5120	Electrochemical Methods of Analysis	3
CHEM 5130	Analytical Molecular and Atomic Spectroscopy	3
CHEM 5160	Analytical Separations	3
CHEM 5310	Organic Synthesis I	3
CHEM 5320	Organic Synthesis II	2
CHEM 5370	Physical Organic Chemistry I	3
CHEM 5380	Physical Organic Chemistry II	3
CHEM 5490	Nuclear Magnetic Resonance Spectroscopy	3
CHEM 5610	Fundamentals of Quantum Mechanics	4
CHEM 5620	Fundamentals of Atomic and Molecular Quantum Mechanics	3
CHEM 5630	Statistical Mechanics	3
CHEM 5640	Molecular Spectroscopy and Structure	3
CHEM 5710	Solid-State Chemistry	2
CHEM 5720	Spectrometric Identification of Organic Compounds	3
CHEM 5730	Nanochemistry	2
CHEM 5740	Organometallic Chemistry of the Transition Metals	2
CHEM 5750	Diffraction and Crystal Structure	3
CHEM 5760	Surface Chemistry	3

CHEM 5770	Mass Spectrometry	3
CHEM 5780	Chemical Kinetics and Mechanisms	2
CHEM 5800	Introduction to Computational Quantum Chemistry	3
CHEM 5830	Chemical Group Theory	1
BBMB 3010	Survey of Biochemistry	3
BBMB 3030	General Biochemistry	3
BBMB 3120	Experimental Research Skills in Biochemistry	2
BBMB 3160	Principles of Biochemistry	3
BBMB 4040	Biochemistry I	3
BBMB 4050	Biochemistry II	3
BBMB 4110	Techniques in Biochemical Research	4
BBMB 4610	Molecular Biophysics	2
BBMB 4900	Independent Study	1-3
BBMB 4990	Undergraduate Research	1-5
BBMB 5040	Amino Acids and Proteins	2
BBMB 5050	Bioenergetics and Metabolism	2
BBMB 5060	Membrane Biochemistry	2
BBMB 5070	Biochemistry of Nucleic Acids	2
BBMB 5610	Molecular Biophysics	2
BBMB 5610L	Laboratory in Molecular Biophysics	2
BBMB 5690	Structural Bioinformatics	3
FSHN 3110	Food Chemistry	3
FSHN 4100	Food Analysis	3
GEOL 4190	Aquatic Geochemistry	3
GEOL 3150	Mineralogy and Earth Materials	3
GEOL 5190	Aquatic Geochemistry	3
ENSCI 4190	Aquatic Geochemistry	3
ENSCI 5190	Aquatic Geochemistry	3
LAS 4990	Internship	1-4
ANS 3190	Animal Nutrition	3

All students majoring in Chemistry are strongly encouraged to take CHEM 3990 and CHEM 5500

LAS 4990 counts only if the internship occurred in a chemistry related workplace.

Only one of BBMB 3010, BBMB 3030, BBMB 3160, the [BBMB 4040 - BBMB 4050] sequence, or the [BBMB 5040 – BBMB 5060] sequence may count as Advanced Chemistry Elective.

Only one of BBMB 4610 or BBMB 5610 may count as Advanced Chemistry Elective.

Chemistry, B.A.

Undergraduate students seeking the Bachelor of Arts (B.A.) degree in Chemistry must complete 45-47 credits in required chemistry, math, and physics classes and have the following courses in their degree programs as minimum requirements. The chemistry requirement includes all Chemistry B.A. Core Courses and the Upper-level Chemistry Choice courses.

Chemistry Core Courses (B.A.)

CHEM 1770	General Chemistry I	4
CHEM 1770L	Laboratory in General Chemistry I	1
CHEM 1780	General Chemistry II	3
CHEM 2110	Quantitative and Environmental Analysis	2
CHEM 2110L	Quantitative and Environmental Analysis Laboratory	2
CHEM 2310	Elementary Organic Chemistry	3
CHEM 2310L	Laboratory in Elementary Organic Chemistry	1
CHEM 3010	Inorganic Chemistry	2
CHEM 3210L	Laboratory in Physical Chemistry	2
CHEM 3250	Chemical Thermodynamics	3
or CHEM 3240	Introductory Quantum Mechanics	

[CHEM 1670/CHEM 1670L (5 cr.)] may be substituted for CHEM 1770/CHEM 1770L (5 cr.) only if the Chemistry major has been declared after receiving credit for [CHEM 1670/CHEM 1670L]

[CHEM 2010/CHEM 2010L (6 cr.)] may be substituted for [CHEM 1770/CHEM 1770N/CHEM 178 (8 cr.)] ^{1,F}

Students choosing Physical Chemistry as Upper-Level Choice must include both CHEM 3240 and CHEM 3250

Students choosing Organic Chemistry as Upper-Level Choice must substitute [CHEM 3310/CHEM 3330L] for [CHEM 2310/CHEM 2310L].

Total Credits **23**

¹ Advanced high school chemistry and strong algebra skills are necessary for success in CHEM 2010. Math ACT of 24 or greater is strongly recommended.

^F Offered only in Fall

^S Offered only in Spring

Upper-level Chemistry Choice (B.A.): choice of one discipline (both courses in the discipline)

Analytical Chemistry

CHEM 3160	Instrumental Methods of Chemical Analysis	2
CHEM 3160L	Instrumental Analysis Laboratory	2

Inorganic Chemistry

CHEM 4010L	Inorganic Chemistry Laboratory	1
CHEM 4020	Advanced Inorganic Chemistry	3

Physical Chemistry

CHEM 3240	Introductory Quantum Mechanics	3
CHEM 3250	Chemical Thermodynamics	3
CHEM 3260	Chemical Kinetics	1

Organic Chemistry

CHEM 3310 & CHEM 3330L	Organic Chemistry I and Laboratory in Organic Chemistry I (for Chemistry and Biochemistry Majors)	5
CHEM 3320 & CHEM 3340L	Organic Chemistry II and Laboratory in Organic Chemistry II (for Chemistry and Biochemistry Majors)	5

Students choosing Physical Chemistry as Upper-Level Choice must include both CHEM 3240 and CHEM 3250

Students choosing Organic Chemistry as Upper-Level Choice must substitute [CHEM 3310/CHEM 3330L] for [CHEM 2310/CHEM 2310L] within the Chemistry Core Courses.

[CHEM 3310L/CHEM 3320L] may be substituted for [CHEM 3330L/CHEM 3340L].

Additional Chemistry Courses (B.A.)

CHEM 1010A	Chemistry Learning Community Orientation: On-Campus Orientation ^F	1
CHEM 1010B	Chemistry Learning Community Orientation: Professional Development Opportunities ^S	1
CHEM 1100	Cutting-Edge Chemistry: Research and Career Opportunities ^F	1

CHEM 1010A and CHEM 1010B are required for students in the Chemistry Learning Community

All students majoring in Chemistry are strongly encouraged to take CHEM 1100

^F Offered only in Fall

^S Offered only in Spring

Math and Physics Core Courses (B.A.)

MATH 1650	Calculus I	4
MATH 1660	Calculus II	4
PHYS 2310	Introduction to Classical Physics I	4
PHYS 2310L	Introduction to Classical Physics I Laboratory	1
PHYS 2320	Introduction to Classical Physics II	4
PHYS 2320L	Introduction to Classical Physics II Laboratory	1

[PHYS 1310/1310L and PHYS1320/1320L] may be substituted for [PHYS 2310/2310L and 2320/2320L]; however, [PHYS 2310/2310L and PHYS 2320/2320L] are highly recommended

Total Credits **18**

Teacher Preparation Focus

Chemistry majors seeking certification to teach Chemistry in secondary schools can follow the requirements of either the B.S. or the B.A. degree in Chemistry. In addition, they must take the complementary coursework listed below and all of the requirements of the Secondary Major in Education. (<https://catalog.iastate.edu/collegeofhumansciences/educationsecondary/#curriculumtext>)

Complementary Coursework

EDUC 4180	Secondary Science Methods I	3
EDUC 4190	Secondary Science Methods II	3
PSYCH 2300	Developmental Psychology	3

One course in Biology (BIOL 2110 & BIOL 2110L recommended)

One course in American History or Government

Chemistry Career Pathways

Students who are planning to attend graduate school in Chemistry or become chemistry professionals are strongly advised to pursue a B.S. in Chemistry.

All other students who want to become teachers, attend professional school (e.g., medical, veterinarian, law) or want to pursue non-traditional chemistry careers such as Forensic Science Technician, Environmental Policy Analyst, Science Communicator, Technical Writer, Laboratory Technician, or other, are advised to pursue a B.A. in Chemistry and complete one of several completion pathways for the B.A. in Chemistry major. This coursework can help a Chemistry graduate establish a clear strength in several fields of application of chemistry. These pathways also make additional majors or minors in other disciplines extremely feasible.

Contact the Chemistry advisor for further information concerning programs of study in Chemistry. Sample Chemistry career pathways are listed at the Chemistry Department web site (<https://www.chem.iastate.edu/undergraduate-student-info/>).

University and/or College Requirements

As majors in the College of Liberal Arts and Sciences, Chemistry students must meet College of Liberal Arts and Sciences (<http://catalog.iastate.edu/collegeofliberalartsandsciences/#lascollegerequirementstext>) and University-wide requirements (<http://catalog.iastate.edu/collegescurricula/>) for graduation, in addition to those stated above for the major.

LAS majors require a minimum of 120 credits, including a minimum of 45 credits at the 3000/4000 level. You must also complete the LAS world language requirement and career proficiency requirement.

Communication Proficiency requirement: The College of Liberal Arts and Sciences aligns with the University Communication Proficiency Grade Requirement (<https://catalog.iastate.edu/academics/#communicationproficiencytext>). At a minimum, students must demonstrate their communication proficiency by earning a grade of C or better in ENGL 2500. The department requires a grade of C- or better in ENGL 3140.

Required General Education electives include a minimum of 12 credits from Group I (Arts and Humanities) and a minimum of 9 credits from Group III (Social Sciences).

Additionally, all ISU majors must complete a three-credit course in U.S. Cultures and Communities and a three-credit course in International Perspectives. These courses may also be used to meet Group I and III Requirements. Discuss with your advisor how the two courses that you select can be applied to your graduation plan.

Lists of approved courses are available online: LAS General Education Approved Course List (<https://las.iastate.edu/academics/general-education/general-education-approved-course-list/>)

Arts and Humanities	12
Social Sciences	9
LAS 2030 Professional Career Preparation	1
LIB 1600 Introduction to College Level Research	1
ENGL 1500 Critical Thinking and Communication	3
ENGL 2500 Written, Oral, Visual, and Electronic Composition	3
ENGL 3140 Technical Communication	3
World Language (First Semester)	4
World Language (Second Semester)	4
Completion of three years of a world language in high school fulfills this requirement.	
Total Credits	40

Chemistry, B.S

Below is a suggested pathway for new majors. This plan is an example only; students should consult their advisers for their individualized needs. See the Chemistry webpage (<https://www.chem.iastate.edu/>) for additional suggested pathways geared towards specific careers.

Freshman

Fall	Credits Spring	Credits
CHEM 1770	4 CHEM 1780	3
CHEM 1770N	1 CHEM 1010B (recommended)	1
CHEM 1010A (recommended)	1 MATH 1660	4
MATH 1650	4 ENGL 2500	3
ENGL 1500	3 General Education/Electives	3
LIB 1600	1	
General Education/Electives	3	
	17	14

Sophomore

Fall	Credits Spring	Credits
CHEM 3310	3 CHEM 3320	3
CHEM 3330L	2 CHEM 3340L	2
CHEM 1100 (recommended)	1 CHEM 2110	2
MATH 2650	4 CHEM 2110L	2
PHYS 2310	4 PHYS 2320	4
PHYS 2310L	1 PHYS 2320L	1
General Education/Electives	3 LAS 2030	1
	18	15

Junior

Fall	Credits Spring	Credits
CHEM 3240	3 CHEM 3250	3
CHEM 3160	2 CHEM 3220L	3
CHEM 3160L	2 CHEM 3010	2
World Language (First Semester)	4 CHEM 5500 (recommended)	1
General Education/Electives	3 World Language (Second Semester)	4
	General Education/Electives	3
	14	16

Senior

Fall	Credits Spring	Credits
CHEM 4020	3 CHEM 4010L	1
ENGL 3140	3 Advanced Chemistry Choice 2	2-3
Advanced Chemistry Choice 1	2-3 BBMB 3010 (recommended)	3
General Education/Electives	6 CHEM 3990 (recommended)	1-6

General Education/Electives	5
14-15	12-18
Total Credits: 120-127	

Chemistry, B.A.

Below is a suggested pathway for new majors. This plan is an example only; students should consult their advisers for their individualized needs. See the Chemistry webpage (<https://www.chem.iastate.edu/>) for additional suggested pathways geared towards specific careers.

Freshman

Fall	Credits Spring	Credits
CHEM 1770	4 CHEM 1780	3
CHEM 1770N	1 CHEM 1010B (recommended)	1
CHEM 1010A (recommended)	1 MATH 1660	4
MATH 1650	4 ENGL 2500	3
ENGL 1500	3 General Education/Electives	3
LIB 1600	1	
	14	14

Sophomore

Fall	Credits Spring	Credits
CHEM 2310	3 CHEM 2110	2
CHEM 2310L	1 CHEM 2110L	2
CHEM 1100 (recommended)	1 PHYS 2320	4
PHYS 2310	4 PHYS 2320L	1
PHYS 2310L	1 LAS 2030	1
General Education/Electives	6 Career Path Choice	6
	16	16

Junior

Fall	Credits Spring	Credits
CHEM 3250	3 CHEM 3010	2
Chemistry Upper-Level Choice	4-5 CHEM 3210L	2
World Language (First Semester)	4 CHEM 5500 (recommended)	1
General Education/Electives	3 World Language (Second Semester)	4
Career Path Choice	3 General Education/Electives Career Path Choice	3
	17-18	15

Senior

Fall	Credits Spring	Credits
ENGL 3140	3 CHEM 3990 (recommended)	1-6
General Education/Electives	3 General Education/Electives	3
Career Path Choice	9 Career Path Choice	9
	15	13-18

Total Credits: 120-126

Chemistry, B.A. with Teacher Preparation & Education Secondary Major

Freshman

Fall	Credits Spring	Credits
CHEM 1770	4 CHEM 1780	3
CHEM 1770N	1 CHEM 1780L	1
CHEM 1010A (recommended)	1 CHEM 1010B (recommended)	1
MATH 1650	4 MATH 1660	4
ENGL 1500	3 EDUC 2040	3
LIB 1600	1 EDUC 2190	1
General Education/Electives	3 EDUC 2800L PSYCH 2300 Apply/Accepted to Educator Preparation Program	0.5 3
	17	16.5

Sophomore

Fall	Credits Spring	Credits
CHEM 2310	3 CHEM 2110	2
CHEM 2310L	1 CHEM 2110L	2
CHEM 1100 (recommended)	1 Chemistry Upper-Level Choice	4-5
PHYS 2310	4 PHYS 2320	4
PHYS 2310L	1 PHYS 2320L	1
EDUC 2020	3 LAS 2030	1
EDUC 3330	3 ENGL 2500	3
	16	17-18

Junior

Fall	Credits Spring	Credits
CHEM 3250	3 CHEM 3010	2
World Language (First Semester)	4 CHEM 3210L	2
EDUC 4180	3 World Language (Second Semester)	4
EDUC 3800B	1-2 CHEM 5500	1

EDUC 3950	3	EDUC 4190	3
		EDUC 4800B	2
		SPED 4010	3
		Apply to Student Teach	

14-15 17

Senior

Fall	Credits	Spring	Credits
ENGL 3140	3	EDUC 4170J	8
HIST 2210 or 2220 (encouraged humanities)	3	EDUC 4170J	8
POLS 1110 (encouraged social sciences)	3		
EDUC 4060	3		
General Education/Electives	6		
Apply for Graduation			

18 16

Total Credits: 131.5-133.5

Minor in Chemistry

As per University requirement, the minor must fulfill the following requirements:

- at least 9 credits in courses numbered 2000 or above, of which at least 6 credits are in courses numbered 3000 or above.
- at least 3 credits taken at Iowa State University.
- at least 3 credits that are not used to meet any other department, college, or university requirement.

Credits used to meet the minor requirements may also be used to satisfy the credit requirement for graduation and to meet credit requirements in courses numbered 3000 or above.

Some students may have to exceed the graduation credit requirement set by their college in order to meet the requirements of both the minor and the major. Courses taken for a minor may not be taken on a pass-not pass basis. A student may not receive both an undergraduate major and a minor of the same name.

In addition to University policies governing minors (<https://catalog.iastate.edu/academics/#degreeplanningtext>), LAS minors require at least 6 credits in courses numbered 3000 and above, taken at ISU with a grade of C or higher. Completed minors and certificates will be recorded on the transcript.

The Department offers a Minor in Chemistry which may be earned by completing 15 credits in chemistry courses chosen as follows:

Required Discipline

CHEM 1770 & CHEM 1780 or CHEM 2010	General Chemistry I and General Chemistry II ¹ Advanced General Chemistry	5-7
CHEM 1770L or CHEM 2010L	Laboratory in General Chemistry I Laboratory in Advanced General Chemistry	1
CHEM 1630/L and CHEM 1670/L can be used in place of CHEM 1770/L if the Minor was declared after credit was received.		

Total Credits 6-8

The remaining 7-9 credits can be chosen among the following courses in ANY combination (i.e., courses do not need to be all from the same Chemistry focus area). Some courses might have pre-requisites. Some paths may require more than 15 credits. Students may count only one of these two combinations [CHEM 2310/CHEM 2310L] or [CHEM 3310/CHEM 3310L & CHEM 3320/CHEM 3320L] toward the Minor.

Chemistry Area Options²

Analytical Chemistry

CHEM 2110 & 2110L	Quantitative and Environmental Analysis and Quantitative and Environmental Analysis Laboratory	4
CHEM 3160 & 3160L	Instrumental Methods of Chemical Analysis and Instrumental Analysis Laboratory	4

Organic Chemistry (Students may count one of these organic chemistry combinations toward the Minor)

CHEM 2310 & 2310L	Elementary Organic Chemistry and Laboratory in Elementary Organic Chemistry	4
CHEM 3310 & 3310L & CHEM 3320 & CHEM 3320L	Organic Chemistry I and Laboratory in Organic Chemistry I and Organic Chemistry II and Laboratory in Organic Chemistry II	8

Inorganic Chemistry

CHEM 3010	Inorganic Chemistry	2
CHEM 4020	Advanced Inorganic Chemistry	3
CHEM 4010L	Inorganic Chemistry Laboratory	1

Physical Chemistry

CHEM 3240	Introductory Quantum Mechanics	3
CHEM 3250	Chemical Thermodynamics	3
CHEM 3260	Chemical Kinetics	1
CHEM 3210L or CHEM 3220L	Laboratory in Physical Chemistry Laboratory in Physical Chemistry	2

¹ Chem 1630/L and Chem 1670/L can be used in place of Chem 1770/L if the Minor was declared after credit was received.

² 4000- and 5000-level courses listed on the Chemistry catalog can also be chosen to fulfill the 15-credit requirement for the Minor.

Concurrent Program

The concurrent Chemistry/MBA program is designed to be completed in five academic years, with three years of undergraduate coursework followed by two years of MBA studies alongside the remaining undergraduate courses.

Concurrent students are part of the full-time MBA program. Admission to the concurrent MBA program is competitive. You should apply during your junior year. Successful candidates demonstrate superior academic ability, strong leadership attributes, and clear career focus. Previous business coursework is not a requirement for application.

Internships, co-op experiences, extracurricular, and leadership activities are highly valued in the MBA candidate selection process. However, you are not required to have additional coursework in business in preparing for the MBA.

Admitted students typically have a cumulative GPA of 3.00 or above and have scored 600 or higher on the GMAT exam or GRE equivalent. If you are interested in the concurrent MBA program, you should apply by **March 15** of your junior year to be considered for scholarships and assistantships. The final application deadline is June 1.

Graduate Programs

The Department offers work for the degrees Master of Science and Doctor of Philosophy with majors in Chemistry, Analytical, Inorganic, Organic, and Physical Chemistry. Co-majors may be taken between areas within Chemistry or between one of the areas in Chemistry and another department. Courses in other areas of Chemistry as well as courses in other departments may be used to satisfy the requirement for coursework outside the major field. A Ph.D. student in Chemistry may choose an additional specialty in one of six areas: Materials Chemistry, Industrial Chemistry, Biomolecular Sciences, Chemistry Education, Chemical Instrumentation, and Forensic Chemistry. A minimum of ten credits is required for each additional specialty. A course which counts towards an additional specialty may also count toward the outside course requirement. A minor in Chemistry is available to students in other departments. The Department participates in the interdepartmental major in Toxicology.

The Department of Chemistry requires all graduate students majoring in Chemistry to teach as part of their training for an advanced degree. Prerequisite to major graduate work is the completion of undergraduate work in Chemistry, Mathematics, and Physics substantially equivalent to that required of undergraduate Chemistry majors at this institution.